

Fall 2012

Problem 1 (Fa12). Find the length of the spiral given in polar coordinates by $r = e^\theta$, $-\infty < \theta \leq 0$.

Problem 2 (Fa12). Prove or disprove the following assertion: If $f: \mathbb{R} \rightarrow \mathbb{R}$ has the property that $f([a, b])$ is a bounded closed interval for every $a \leq b$, then f is continuous.

Problem 3 (Fa12, Sp21). Prove the existence of the limit

$$\lim_{n \rightarrow \infty} \left(\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} - \log n \right)$$

Problem 4 (Fa12, Fa17, Fa20). 1. Find the poles and residues of $\frac{1}{z^3 \cos z}$.

2. Show that the integral of the function above over a square contour centered at the origin with side $2\pi N$ tends to zero as the integer N tends to infinity.

3. Find the sum $\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \frac{1}{7^3} + \cdots$

Problem 5 (Fa12). 1. Show that if $|z_0| < 1$ then there is a holomorphic function defined on some neighborhood of the unit disc whose only zero is at z_0 and that has absolute value 1 on the unit circle.

2. Suppose that f is a holomorphic function on the complex plane and is not identically zero. Show that there is a holomorphic function g defined in some open set containing the unit disc such that $|f(z)| = |g(z)|$ whenever $|z| = 1$, and such that g has no zeros in the open unit disc.

Problem 6 (Fa12). If U , V , W are subspaces of a vector space such that any two have intersection zero, prove that

$$\dim(U+V+W) + \dim U + \dim V + \dim W \leq \dim(U+V) + \dim(V+W) + \dim(W+U)$$

and give an example where equality does not hold.

Problem 7 (Fa12). Let I_n denote the $n \times n$ identity matrix, and J_n the $n \times n$ matrix with all entries equal to 1. Determine for which real numbers a the matrix $I_n + aJ_n$ is invertible, and find its inverse.

Problem 8 (Fa12). Let G be a finite abelian group of order n . Suppose m is a squarefree, i.e., not divisible by the square of a prime, positive integer dividing n . Show that G contains an element of order m . Give an example to show that this need not be true if m is not assumed to be squarefree.

Problem 9 (Fa12, Sp21). Does there exist a homomorphism of commutative rings with unit from $\mathbb{Z}[x]/\langle x^2 + 3 \rangle$ to $\mathbb{Z}[x]/\langle x^2 - x + 1 \rangle$? Either exhibit such a homomorphism, or prove that none exists.

Problem 10 (Fa12). Prove that

$$\frac{\pi}{4} = 4 \arctan \frac{1}{5} - \arctan \frac{1}{239}$$

In 1706 John Machin used this formula to calculate π to 100 decimal places. Explain briefly why he did not use the simpler formula $\pi/4 = \arctan 1$.

Problem 11 (Fa12). 1. Find the sum $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$

2. Find the sum $1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \frac{1}{5} - \frac{1}{10} - \frac{1}{12} + \cdots$

Problem 12 (Fa12). Let (X, d_X) and (Y, d_Y) be metric spaces. Suppose that a map $\pi : X \rightarrow Y$ is a *submetry*; this means that for every $x \in X$ and any $r > 0$, the image of the closed r -ball around x is the closed r -ball around $\pi(x)$.

1. Show that π is surjective if X is nonempty.
2. Show that π is continuous.
3. Show that π is open, that is the image of any open subset is open.

Problem 13 (Fa77, Su82, Fa97, Sp01, Fa03, Fa08, Sp11, Fa12, Sp18, Sp21). Evaluate $\int_{-\infty}^{\infty} \frac{dx}{1 + x^{2n}}$, where n is a positive integer.

Problem 14 (Sp11, Fa12). Show that $e^z = z$ has infinitely many complex roots.

Problem 15 (Fa12). Find all eigenvalues and eigenvectors of the linear map $T: \mathbb{C}^n \rightarrow \mathbb{C}^n$ given by $T(x_1, \dots, x_n) = (x_2, x_3, \dots, x_n, x_1)$.

Problem 16 (Fa12). Suppose that A and B are linear transformations of a finite dimensional complex vector space such that $AB - BA = A$. If v is an eigenvector of B with eigenvalue λ , show that Av is zero or an eigenvector of B and find its eigenvalue. Prove that A is nilpotent.

Problem 17 (Fa12). Let R be a commutative ring with identity. Suppose that there is a monic polynomial $p(x) \in R[x]$ such that the ideal $\langle p(x) \rangle \subseteq R[x]$ is maximal. Prove that R is a field.

Problem 18 (Fa12). Let M be a (possibly singular) square matrix over a field $\mathbb{F}$. Let p be the product of the nonzero eigenvalues of M (counted with multiplicities) in some algebraically closed extension $\mathbb{K}$ of $\mathbb{F}$. Prove that $p \in \mathbb{F}$.